

NEWDIGS / Tufts FoCUS — Cell & Gene Therapy

Key concepts for your rare monogenic GT dissertation | Source: newdigs.tuftsmedicalcenter.org | Data through 2024

1. What is a Durable Cell & Gene Therapy (CGT)?

CGTs are **one-time (or very-few-time) treatments** designed to treat or prevent disease by introducing, removing, or altering genetic material within a patient's cells. They are *durable* — unlike chronic medicines taken repeatedly for life.

- Regulated by FDA CBER (Center for Biologics Evaluation and Research)
- Enabling statutes: Public Health Service Act + Federal Food, Drug & Cosmetic Act
- Each product targets a specific gene — the target gene is a key identifier for matching to disease
- All products on the NEWDIGS list have completed the full FDA drug approval process

2. The Two Delivery Approaches: Ex Vivo vs In Vivo

The most important classification to understand is **where** the genetic modification happens.

	EX VIVO	IN VIVO
What it means	Cells taken out of patient → modified in lab → returned	New genetic material delivered directly into the body
Methods	CAR-T, ex vivo gene transfer (lentiviral / retroviral HSC)	Viral vectors (AAV), CRISPR-Cas9, RNAi, naked DNA injection
Process	Complex — cell extraction → lab modification → quality check → reinfusion	Simpler — infusion or direct injection into target organ
Control	Higher — lab can verify modification before it goes back in	Lower — immune response to vector can affect outcome
Best for	Blood/immune cells (accessible & manipulable outside body); haematological diseases	Single-gene disorders affecting internal organs (liver, brain, eye, muscle, lung, heart)
Cost / time	Higher cost, longer process	Lower process complexity
Your dissertation relevance	SCD, beta-thal, ADA-SCID, MLD, ALD — blood/immune rare monogenic diseases	Most relevant for rare monogenic diseases of internal organs — Zolgensma, Luxturna, Hemgenix, etc.

3. Why CGTs Are Disruptive for Health Systems

NEWDIGS focuses on **payment and access challenges**, not biology. These are the key problems:

Challenge	Why it matters
One-time high cost	Standard drug payment models (monthly prescription = monthly payment) don't fit a one-time treatment costing \$1M+. Payers face huge upfront costs for benefits that accrue over years.
Uncertain durability	Long-term efficacy data is limited. Payers don't know if a 'cure' lasts 5 years or a lifetime, making value assessment difficult.
Patient mobility	Patients may change insurer between treatment and when benefits are realised — the insurer who paid for the cure may not be the one who benefits from it.
Specialised treatment centres	CGTs require Centres of Excellence, limiting geographic access. Not all patients can travel to approved centres.
Outcomes tracking	Long-term follow-up data is hard to collect, making value-based contracts difficult to implement.
Provider financial risk	'Buy and bill' models expose hospitals to financial risk if a payer refuses to cover the therapy after administration.

4. Precision Financing Solutions (NEWDIGS proposed models)

Because standard payment models fail for CGTs, NEWDIGS developed several alternative financing approaches:

Model	How it works
Milestone-based contracts	Payer pays in installments tied to patient health outcomes at defined time points. If therapy fails, payments stop.
Performance-based annuities	Price is paid as an annual stream; size depends on whether the patient continues to benefit.

Model	How it works
Payment over time / Installments	Simply spreading the upfront cost — less sophisticated but lowers initial burden.
Reinsurance / Stop-loss	External insurer covers costs above a threshold — protects payers from catastrophic CGT spend.
Risk pools	Multiple payers pool CGT risk, reducing any one payer's exposure.
ORBM (Orphan Reinsurer & Benefit Manager)	Specialist intermediary to manage rare-disease CGT payment across payers.
Warranty	Manufacturer guarantees a refund or rebate if the therapy doesn't work as claimed.
Subscription	Payer pays a flat annual fee for unlimited access to a product across their population.

5. CGT Pipeline Outlook (NEWDIGS estimates, data through Dec 2023)

NEWDIGS/FoCUS estimated **~85 product-indication approvals by 2033**. By ~2026 the following were expected:

- **Oncology:** additional CAR-T therapies for B-cell leukaemia / lymphoma
- **Rare disease:** haematological conditions, more ophthalmic CGTs, neurological conditions
- **Higher-prevalence:** age-related wet macular degeneration, osteoarthritis (knee)

■ **Note:** NEWDIGS FoCUS concluded new research in 2024 — pipeline figures are not being updated. Cross-check against FDA approvals list, ClinicalTrials.gov, and recent review articles.

6. How to Use This for Your Rare Monogenic GT Dissertation

Question	NEWDIGS usefulness
Which CGTs are FDA-approved?	✓ High — maintained approved therapies list (Excel download on site)
What disease does each treat?	✓ High — indication listed per product
In vivo or ex vivo?	✓ High — delivery approach categorised per product
Target gene?	✓ High — gene targeted listed, links to Individual Indication workbook
Rare / orphan?	✓ Useful — orphan status noted
Access / payment challenges?	✓ Very useful — core focus of the site
Disease biology / mechanism?	✗ Not covered — use FDA labels, GeneReviews, OMIM, Orphanet
Most up-to-date global list?	✗ No — cross-check elsewhere (FDA, EMA, recent papers)

Source: newdigs.tuftsmedicalcenter.org/payingforcures — NEWDIGS FoCUS Project (archived 2024). Approved therapies Excel: wp-content/uploads/2025/06/Approved-Cell-and-Gene-Therapies-update-20250605.xlsx